Towards a Seismic Vulnerability Database for Benefit-Cost Analysis considering Retrofitting Interventions

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ABSTRACT

Earthquakes pose significant risks to communities, largely due to vulnerable structures designed prior to the introduction of modern seismic provisions. Retrofitting these structures can significantly reduce seismic vulnerability, and lower potential losses. In this context, it is crucial to refine capacity and vulnerability models to reflect these structural interventions.Currently, the absence of vulnerability models that consider retrofitting interventions presents challenges in accurately evaluating the seismic risk of retrofitted structures. To this end, this study aims to create a database of vulnerability functions that specifically accounts for the impact of retrofitting, distinguishing retrofitted building classes from their original counterparts. Updating models such as the global vulnerability database recently published by the Global Earthquake Model (GEM) Foundation to account for retrofitting will enable a more accurate representation of seismic vulnerability and allow for more precise loss estimations. The outcome of this study can enable risk modelers in performing cost-benefit analysis, one of the most common tools to assess the effectiveness and financial viability of different risk reduction measures.